

The Routine — Balanced PPL×2 @ 290 Revolution

Dr

Generated from the [exercise](#) corpus (58 muscles · 233 exercises · 103 training_variables · 13 set_structures · 46 adversarially-verified claims) against the 290 Revolution equipment load-algebra and Mark's constraint profile. **Not medical advice** — the lower-back priority is supportive strengthening; keep the physio assessment, and stop on any red flag.

Philosophy & schedule

RIR TARGETS (evidence-based): Heavy compounds (squat, deadlift, hinge, fixed-core press/pull) = 2-3 RIR most sets, reserving 0-1 RIR for the very last set only - failure on big barbell lifts is slightly INFERIOR for strength and triples acute neuromuscular fatigue (Robinson/Ruple/Refalo), and on the deadlift it threatens the priority lower-back. Isolation/machine/cable = 0-2 RIR, final set often to failure (cheap fatigue, captures the small hypertrophy premium). At light loads (high-rep cable/calf) train 0-1 RIR so high-threshold units are recruited. Strength-leaning compound work can sit 3-5 RIR and still build strength at matched volume.

PROGRESSION: Double progression - add reps within each exercise's prescribed range, then add load (5-lb barbell steps / next DB) once the top of the range is hit at target RIR. Run ~4-6 week accumulation blocks ramping volume from the low end (~12 sets/muscle) toward the high end (~18-20) and RIR DOWN within the block (e.g. start ~3-4 RIR, finish ~1 RIR) - matched gains at lower average RPE (Martikainen). Then a deload week at ~half volume and 3-6 RIR to shed accumulated fatigue, and restart. The erector/core priority gets the high end of its range and the most consistent weekly load progression because it's painless and trainable - but only while it stays painless (any leg radiation, numbness, give-way, night pain = stop and reassess).

AUTOREGULATION AROUND VARIABLE SPORTS (the lifting is self-sufficient; sports are NOT counted as training volume): The plan is built to stand alone, so on heavy-sport-overlap weeks you AUTOREGULATE the lifting down rather than leaning on the sport for stimulus. Levers, in order: (1) raise RIR by 1-2 on the affected session, (2) drop the last 1-2 accessory sets, (3) on a Tue (pre-kickboxing) or Sat (post-krav) keep big lifts at 2-3 RIR and never to failure. Pull is deliberately scheduled on the kickboxing day (Tue) and the lighter chest-supported Pull after krav (Sat) so leg/push power is fresh for the sport; the heavy fatiguing hinge is isolated to Wed -> Thu rest. If a sport session is unusually brutal, the missing lifting volume is made up later in the week or simply skipped - never added on top into overreaching.

CONSTRAINT INTEGRATION (always on): Shoulder painless-crepitus -> neutral-grip DB / landmine / incline / Smith / machine presses are the press anchors; NO heavy strict barbell overhead press; rotator-cuff + scapular work every Push/Pull. Wrist tendinopathy -> neutral-grip variants, wrist wraps, cable/machine where the wrist stays neutral. Left pinky (temporary) -> FTS cable wrist cuffs / lifting straps / machine-loaded variants on every grip-loaded pull, hinge, and carry; defer crush-grip (gripper/pinch/dead-hang) until it settles. TMJ -> jaw relaxed / mouthguard, no hard clench on any heavy or valsalva set.

EXERCISE-SELECTION LOGIC (from the verified claims): Picks are justified by ROM/muscle-length, stimulus-to-fatigue ratio, and constraint-fit - NOT by EMG/pump 'targeting' (acute EMG and pump do NOT predict regional growth - refuted). Claims I actively USED: standing (gastroc) vs seated (soleus) calf raises; seated leg curl for the hip-extensor

hamstring heads; hip-hinge vs leg-curl hamstring split (program BOTH); reclined/extended-hip leg extension for rectus femoris; the modest lengthened-position growth edge (incline curl, lat-prayer, stretch-loaded raises, deep squat). Claims I deliberately did NOT lean on: grip steers lat region, hip-thrust > squat for glute growth, behind-neck press, incline-maximizes-or-combine-for-upper-pec, leg-press/squat-stance muscle reallocation - so the hip thrust is in for the short-length emphasis it adds (not as a squat replacement), and multiple press/pull angles are in for coverage and joint variety (not as proven region-specific growth). Advanced techniques (drop sets, rest-pause, myo-reps, supersets, pyramids) are EQUIVALENT to straight sets at matched volume, so they appear ONLY on the pump days (Fri Push, Sat Pull) purely for time-efficiency and fatigue management - never as a way to claim extra growth. Frequency is the volume vehicle (2x/muscle via PPLx2 satisfies the $\geq 2x$ floor; beyond that it's logistics), and hypertrophy is load-independent from ~30-85% near failure, which is exactly why each muscle's two weekly sessions are split into one heavier/lower-rep and one lighter/higher-rep pass."

Day	Session	Targets	Anchor(s)
Sun - LEGS (moderate, quad-biased)	Legs	quads, glutes, hamstrings, adductors, calves, core	Fixed-core: quad-biased squat to deep ROM (High-Bar / Front / Smith Squat), 3-4 x 5-10 @ 1
Mon - PUSH (heavy)	Push	chest, front delt, lateral delt, triceps	Fixed-core: heavy shoulder-friendly press - Incline/Flat Dumbbell or Machine/Smith Press,
Tue - PULL (controlled, before kickboxing PM)	Pull	lats, upper/mid back, rear delt, biceps, forearms, erectors (carries)	Fixed-core: heavy vertical pull (Weighted/Neutral-grip Pull-Up or Iso-Lateral Pulldown) 5-
Wed - LEGS (heavy hinge - > Thu REST)	Legs	hamstrings, glutes, erectors (PRIORITY), quads, calves, core	Fixed-core: heavy hip-hinge - Conventional or Deficit Deadlift (erector + glute + ham PRIM
Thu - REST	Rest		Full recovery (optional easy mobility / hip-flexor + T-spine, light walking)
Fri - PUSH (pump / metabolite)	Push	chest, lateral delt, triceps, front delt, rotator cuff	Fixed-core (lighter): Machine/Smith Incline or Seated DB Press, 3-4 x 10-15 @ 1-2 RIR - mo
Sat - PULL (lighter, after krav AM)	Pull	lats, upper/mid back, rear delt, biceps, forearms, erectors (carries)	Fixed-core (lighter): Chest-Supported / Iso-Lateral Row 10-15 @ 1-2 RIR (low lower-back de

Weekly volume (balanced; lower-back & core prioritized to bring up the weak point)

Muscle	Weekly sets	Note
Chest	11 direct (Mon 6 heavy press+flye, Fri 5 pump press+crossover)	2x/wk. Anchors are neutral-grip/incline DB + machine/Smith press (shoulder-crepitus friendly), not heavy strict barbell. Heavy day 5-10 reps @1-2 RIR on fixed-core press; pump day 10-25 reps @0-2 RIR on machine/cable with constant tension. Flat vs incline chosen for ROM/comfort, NOT to 'target' a region (incline upper-pec growth is disputed; combining to maximize is refuted).
Back - lats (vertical pull)	9 (Tue 5, Sat 4)	2x/wk vertical pull. Grip orientation/width chosen for joint comfort and stretch ROM, NOT lat-region targeting (refuted). Heavy day = weighted/neutral pull-up or iso-lateral pulldown 6-12; pump day = single-arm pulldown + lat-prayer/straight-arm for lengthened bias. Straps/FTS cuffs bypass left-pinky grip.
Back - upper/mid (horizontal row)	9 (Tue 5, Sat 4)	2x/wk row. Both horizontal rows and vertical pulls programmed for complete back coverage (the row-vs-pulldown split is a sensible heuristic, not proven differential growth). Chest-supported/Smith rows on the Pull days that follow leg work keep lower-back fresh; free barbell row used Tue with straps.

Muscle	Weekly sets	Note
Back - erector spinae (LOWER-BACK PRIORITY)	~14-16 fractional (high end on purpose)	WEAK-POINT BUILD. No 45/GHD/back-ext machine exists, so erectors are loaded as PRIMARY via conventional/deficit deadlift + good-morning (Wed), heavy-SECONDARY via RDL/SLDL and all barbell rows, and isometrically via farmer's/suitcase carries. Painless = progress. Heavy hinge sits Wed before Thu rest; kept off Tue/Sat sport days. Trained 3-4x/wk across hinges+rows+carries to accumulate the high volume safely.
Core (rectus/oblique s/TVA/QL - PRIORITY)	12-14 (Sun 3, Wed 3, plus anti-rotation/carry on push/pull days)	WEAK-POINT BUILD, trained ~4-5x/wk in short doses. Spans all three core functions: spinal-flexion (cable/weighted crunch, hanging leg raise - loaded, progressible, 8-20 reps @0-2 RIR), anti-extension (ab-wheel rollout, weighted plank), anti-lateral-flexion/anti-rotation (suitcase carry, cable/DB side bend). Bracing core + erectors together is the functional-strength + running-tightness fix.
Quads	~16 fractional (Sun 5 squat-bias, Wed 3, +indirect from hinges/lunges/calf-press patterns)	2x/wk, legs FULLY trained. Sun = quad-biased squat (high-bar/front/Smith) to deep ROM (deep squat supported for leg growth) 5-10 @1-2 RIR + leg extension. Reclined/extended-hip leg extension added to bias rectus femoris (a supported claim). Wed quads are accessory to the hinge.
Hamstrings	~14 (Wed 5 hinge+curl, Sun 2 curl, +indirect deadlift)	2x/wk via BOTH movement patterns (supported hamstring split): hip-hinge (RDL/SLDL/good-morning - hip-extensor heads) on Wed, and a knee-flexion leg curl - SEATED prioritized (grows hip-extensor heads more, supported) plus lying for the short head. Wed heavy hinge 5-10 @1-3 RIR; curls 12-30 @0-2 RIR.
Glutes	~14 (Sun 3, Wed 4 +heavy indirect from all hinges/squats/lunges)	2x/wk. Complementary length coverage: long-length via squat + RDL/deficit hinge (Wed), short-length via hip thrust/glute bridge (machine/Smith/barbell). Hip thrust is included for the short-length emphasis it ADDS, not as superior to squat for growth (that claim is refuted). Abduction (cable/machine) for medius/minimus.
Adductors	~6-8 fractional (Sun 2 direct + heavy indirect from deep squats/sumo/lunges)	Adductor magnus is a major hip extensor at squat/hinge depth, so most stimulus is indirect; direct work = Copenhagen plank (also anti-lateral-flexion core) and hip-adduction machine/cable on Sun. Deep/wide stance loads them further.
Calves	~9-10 (Sun 3, Wed 3, optional Push-day 3; +running indirect)	Train BOTH heads by knee angle (supported): standing/knee-straight raises for gastrocnemius (incl. lengthened-partial bottom range), seated/leg-press bent-knee raise for soleus. 10-30 reps @0-1 RIR. Single-leg ankle-stabilizer raise feeds tib-post/fibularis for functional ankle robustness.
Shoulders - lateral delt	8 (Mon 4, Fri 4)	2x/wk, the main delt driver since pressing is shoulder-conservative. Cable/cuff/machine lateral raises (grip-bypass, loaded-stretch bottom) 12-30 @0-2 RIR. Cable chosen for tension profile/comfort; not claimed superior to DB for growth (refuted).
Shoulders - rear delt	~9-12 (Tue 3, Fri 3, Sat 3 + indirect from all rows)	Trained with Pull and as Push-day antagonist filler. Face pulls + rear-delt flies/cuff flies + high-row path. Doubles as scapular-health/posture work for the desk-and-running lifter.
Shoulders - front delt	~12-14 fractional (4 direct + heavy indirect from all pressing)	Mostly indirect from chest/overhead pressing; minimal direct front raise needed. Overhead pressing done as shoulder-friendly neutral-grip DB / landmine / Smith / machine seated press (no heavy strict barbell OHP per crepitus constraint).
Rotator cuff (infra/teres minor/subscap/supraspin)	~4-6 (Fri 2, woven into Pull warm-ups)	Prehab for the crepitus shoulder: cable/DB external & internal rotation, full-can raise. Light 12-20 @1-3 RIR. Not hypertrophy volume - joint-stability insurance that lets the presses stay painless.
Arms - biceps	~14-16 fractional (Tue 4 + Sat 4 direct, heavy indirect from chins/rows)	2x/wk on Pull. Lengthened-bias curl (incline DB) prioritized over peak-contraction (supported modest edge). Heavy EZ/DB curl 5-12 @1-3 RIR Tue; cable/spider pump 10-30 @0-2 RIR Sat. Cuffs/straps for pinky on any hanging load.

Muscle	Weekly sets	Note
Arms - triceps	~15-18 fractional (Mon 5 + Fri 4 direct, heavy indirect from all presses)	2x/wk on Push - and triceps specifically benefit from >20 total sets (synergist), which the indirect pressing supplies. Heavy day = close-grip/JM/overhead EZ extension 5-12 @1-2 RIR; pump day = rope pushdown + overhead cable 15-30 @0-2 RIR (long-head stretch).
Forearms / grip	~5-6 (Tue 2, Sat 3)	Neutral-grip + wrap friendly (wrist tendinopathy). Wrist curl/reverse curl + carries; reverse curl & hammer for brachioradialis. Note: left-pinky crush-grip work (gripper/pinch/dead-hang) is deferred while the pinky is symptomatic - use FTS cuffs/straps to keep training the rest.

The 6 sessions

Sun — LEGS A — Moderate, Quad-Biased (strength-leaning squat anchor; protects the heavy Wed hinge)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
1	High-Bar Back Squat	4 × 6-8	2 (last set 1)	180-240	Olympic bar (5-lb plate steps, ~570 lb ceiling). Start ~185 lb for 4x8 @2 RIR. Double-progress: add reps to 8 across all sets at 2 RIR, then +5-10 lb and drop back to 6. This is the quad/strength anchor of the week — keep a 2-RIR buffer (only the last set to 1) so you are not pre-fatigued for Wed's heavy deadlift.	straight-sets / double-progression (low fatigue, best for the strength-leaning compound; RPT not used here so total CNS cost stays moderate before Wed)
2	Romanian Deadlift (RDL)	3 × 8	2-3	150-180	Olympic bar with straps. Start ~155-185 lb for 3x8 @2-3 RIR. Progress reps then +5-10 lb. Kept MODERATE on purpose (2-3 RIR, mid load) so the hamstring/glute/erector hinge stimulus is there twice a week without pre-cooking the lower back before the heavy Wed pull.	straight-sets / double-progression (spare the spine on the hinge per Helms — chase failure on the isolation work later, not here)
3	Leg Extension (Reclined / Extended-Hip, Rectus-Femoris Bias)	3 × 10-12	1 (last set 0)	90-120	Precor selectorized leg-extension stack. Recline the seat back / extend the hip as far as the pad allows. Start at a load you hit 12 reps with at 1 RIR; add a plate when 3x12 @1 RIR is clean. Cheap fatigue here, so the last set goes to true failure.	straight-sets / double-progression; final set as a brief drop (knock the pin down 1-2 plates, push to failure again) for time-efficient extra reps — equivalence technique, not 'more growth'
4	Seated Leg Curl	3 × 12-15	1-2 (last set 0)	90-120	Precor selectorized seated leg-curl stack. Start at a load for 15 reps @2 RIR; double-progress reps to 15 then add a plate. The seated (hip-flexed) position trains the hip-extensor hamstring heads at long length — the directly-measured higher-growth curl variant.	straight-sets / double-progression; can be antagonist-superset with the leg extension (order 3) to compress time — agonist/antagonist pairing is time-efficient with no growth penalty
5	Smith Machine Hip Thrust	3 × 10-15	1-2 (last set 0-1)	90-120	Smith machine (25-lb effective bar + plates, ~550 lb ceiling). Start ~135-185 lb across the hips for 3x12 @1-2 RIR; add 10-20 lb when the top of the range is clean. Shoulders on a bench, full hip extension, pause/squeeze at lockout.	straight-sets / double-progression (constant short-length tension; Smith fixes the bar path so you can push close to failure safely)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
6	Smith Machine Standing Calf Raise	3 × 8-12	1-2 (last set 0-1)	75- 90	Smith machine standing on a plate/block for full stretch (no dedicated calf machine exists at 290). Start ~135-185 lb for 3x10 @1-2 RIR; add 10 lb when 3x12 is clean. This is the heavier, lower-rep gastroc pass of the week (Wed gets the higher-rep soleus pass).	straight-sets / double-progression; emphasize a 2-3s loaded stretch and full lengthened-range bottom partials on the last set (lengthened bias is a reasonable add-on for the calf)
7	Hip Adduction Machine	2 × 15- 20	0-1	75- 90	Precor selectorized adduction stack. Open the pads to a deep abducted STRETCH, start at a load for 18 reps @1 RIR; add a plate when 2x20 is clean. Most adductor magnus stimulus already comes indirectly from the deep squat — this is the direct top-up.	straight-sets to 0-1 RIR (cheap isolation fatigue); optionally superset with a Copenhagen plank (8-15/side) for an eccentric/lengthened adductor + anti-lateral-flexion core two-for-one
8	Cable Crunch (Kneeling Rope Crunch)	3 × 12- 15	1-2 (last set 0)	60- 90	FTS / cable stack with rope. Start at a load for 15 reps @2 RIR; this is loadable and progressible so double-progress reps then weight. Core is a PRIORITY weak-point — load it like a muscle, not endurance fluff.	straight-sets / double-progression; finish the session by giant-setting a Suitcase Carry (20-40s/side) as a time-cheap erector/QL anti-lateral-flexion + grip finisher (carries load the priority lower-back isometrically)

- **High-Bar Back Squat** — *cue*: Squat to deep ROM (at/below parallel) for full quad length and knee-flexion ROM. Brace, knees track over mid-foot, sit down (knee-dominant) rather than sit back to keep the quads loaded through the longest range. Drive the floor away out of the hole. · *accommodation*: TMJ: keep jaw relaxed / use a mouthguard on the top sets — no hard clench during the valsava brace. Bar on traps (no overhead/shoulder-press demand) so the painless-crepitus shoulder and quiet wrist are unloaded. No grip-limited hanging load, so the left pinky is a non-issue. · *cites*: [exercise.claim.ml-deep-squat-more-leg-growth](#), [exercise.claim.ml-knee-dominant-squat-quad-bias](#), [exercise.claim.17-failure-required-to-maximize-growth-and-strength](#), [exercise.claim.18-zero-rir-beats-low-rir-closer-is-more-growth](#), [exercise.tech.straight-sets](#), [exercise.constraint.mark.tmj](#)
- **Romanian Deadlift (RDL)** — *cue*: Soft knees, push hips straight back, feel the hamstring stretch, lower to mid-shin (long muscle length) and hard glute lockout. Flat neutral spine throughout — this is erector-endurance work too. Stop the set if bar speed or back position degrades. · *accommodation*: Lower-back PRIORITY build: painless = progress weekly; STOP on any leg radiation, numbness, give-way or night pain. Left pinky: lifting straps remove the grip limiter. Wrist tendinopathy: straps + neutral wrist keep it quiet. TMJ: relaxed jaw on the brace. · *cites*: [exercise.claim.ml-rdl-hip-dominant-vs-leg-curl-knee-dominant-hamstring](#), [exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated](#), [exercise.constraint.mark.lower-back](#), [exercise.constraint.mark.left-pinky](#), [exercise.constraint.mark.right-wrist-tendonitis](#)
- **Leg Extension (Reclined / Extended-Hip, Rectus-Femoris Bias)** — *cue*: Extended-hip seating lengthens the biarticular rectus femoris to bias its growth without costing vastus lateralis. Full lockout, controlled lengthened phase, no hip thrusting off the pad. · *accommodation*: Machine, no grip/wrist/shoulder demand — fully constraint-friendly. Pinky/wrist/shoulder all unloaded. · *cites*: [exercise.claim.ml-leg-extension-extended-hip-rectus-femoris](#), [exercise.claim.5-rectus-femoris-active-insufficiency-distal-untrainable](#), [exercise.tech.drop-set](#), [exercise.claim.8-drop-sets-more-muscle](#)
- **Seated Leg Curl** — *cue*: Hips fixed against the pad, full knee flexion squeeze, controlled lengthening. Drive the heels under, don't let the hips lift to cheat. · *accommodation*: Machine — grip/wrist/shoulder all bypassed. No pinky concern. · *cites*: [exercise.claim.ml-seated-vs-lying-leg-curl-length-bias](#), [exercise.claim.ml-rdl-hip-dominant-vs-leg-curl-knee-dominant-hamstring](#), [exercise.tech.antagonist-superset](#), [exercise.claim.11-supersets-compromise-hypertrophy](#)
- **Smith Machine Hip Thrust** — *cue*: This is the SHORT-LENGTH glute complement to the squat/RDL's long-length loading — it is here for the length coverage it ADDS, not because it out-grows the squat (it does not). Posterior pelvic tilt at the top, chin tucked, ribs down. Use a pad across the hips. · *accommodation*: Smith bar across the hips, no grip/hang demand —

pinky and wrist unloaded. No overhead/shoulder loading. TMJ: relaxed jaw at lockout. · *cites*: exercise.claim.ml-hip-thrust-short-length-squat-long-length-glute, exercise.claim.ml-hip-thrust-vs-squat-glute-growth, exercise.claim.4-hip-thrust-superior-glute-growth, exercise.tech.straight-sets

- **Smith Machine Standing Calf Raise** — *cue*: Knee straight = gastrocnemius bias (the directly-measured growth result). Big stretch at the bottom, full plantarflexion squeeze at the top, no bouncing. · *accommodation*: Smith bar on traps/shoulders — no grip/hang load, pinky safe. Wrist neutral. TMJ: relaxed jaw if loading heavy. · *cites*: exercise.claim.2-standing-vs-seated-calf-raise-gastroc, exercise.claim.ml-standing-vs-seated-calf-gastroc-soleus, exercise.claim.10-lengthened-partials-more-than-full-rom, exercise.tech.lengthened-partials
- **Hip Adduction Machine** — *cue*: Full abducted stretch each rep, controlled return — long muscle length is where the adductor responds. If pairing Copenhagen, it doubles as lateral-chain core. · *accommodation*: Machine — no grip/wrist/shoulder demand. Copenhagen (if supersetted) is bodyweight, also constraint-free. · *cites*: exercise.claim.ml-squat-stance-quad-vs-glute-bias, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, weekly_sets
- **Cable Crunch (Kneeling Rope Crunch)** — *cue*: Crunch the rib cage toward the pelvis (spinal flexion), don't just hip-hinge. On the carry finisher: stay tall, resist the side-bend, brace — this is anti-lateral-flexion and erector endurance for the running-tightness fix. · *accommodation*: Cable crunch: rope held at the head, minimal grip — pinky fine. Suitcase carry finisher: load via DB with a lifting strap or FTS cuff so the left pinky is bypassed; keep load moderate and wrist neutral. · *cites*: exercise.constraint.mark.lower-back, exercise.constraint.mark.left-pinky, exercise.tech.giant-set, weekly_sets

Mon — PUSH A — Heavy / Strength (chest, delts, triceps)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
1	Incline Dumbbell Bench Press	4 × 6-8	2 RIR sets 1-3, 1 RIR last set	180	Start the heavy press while fresh. ~30-deg incline. Begin 60s x6-8 (the 60-lb pair). Double-progress: when you hit 8 reps at target RIR on all 4 sets, jump to the next pair (65 -> 70 -> 75). DB caps at 75/hand — when 75s x8 feel like 2+ RIR, switch the heavy-press slot to Smith Incline Press (loadable past the DB ceiling) and demote DBs to the pump-volume role. ~30s drop to set the dumbbells down each set; use knee-kickup to get them up.	straight-double-progression
2	Smith Machine Flat Bench Press	2 × 6-10	2 RIR, then 1 RIR back-off	150	Second press, the angle you didn't anchor with (flat). Fixed bar path lets you load past the DB ceiling and stay safe near failure solo. Smith effective bar ~25 lb; start ~135 effective (25 bar + 2x55) x8 @2 RIR. RPT: top set heaviest, then strip ~10-15% for the back-off set to chase a couple more reps. Add 5-10 lb when you clear 10 reps at 2 RIR.	RPT (reverse-pyramid: heavy top set first, ~10-15% drop for the back-off) — for time-efficiency, NOT extra growth
3	Close-Grip Bench Press	3 × 6-10	1-2 RIR	150	Heavy triceps compound (medial/lateral heads + pressing carryover). Use the Smith for a solo-safe fixed path. Grip ~shoulder-width (not narrower — protects the wrist). Start ~115 effective x8 @2 RIR, add 5 lb when you clear 10 at 1-2 RIR. This is the heaviest triceps stimulus of the week and feeds the >20-set triceps total via indirect pressing volume.	straight-double-progression
4	Machine Lateral Raise	4 × 12-15	1 RIR, last set 0 RIR	90	The main delt driver on Push (pressing is deliberately shoulder-conservative, so direct lateral work carries side-delt growth). Selectorized stack (Precor) — grip- and core-free. Start a load you can clean for 12-15 with 1 RIR; double-progress reps to 15 then add one plate. Even on the heavy day, laterals stay 12-15 (the joint hates low-rep heavy abduction).	straight-double-progression (last set to failure — cheap isolation fatigue)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
5	Overhead Barbell / EZ-Bar Triceps Extension	2 × 10-12	1-2 RIR	120	Long-head stretch bias (overhead = long head lengthened) to complement the elbow-at-side close-grip work. Fixed/EZ bar 20-60 lb in 10-lb steps, or load an Olympic EZ. Start ~50-60 lb x10-12 @2 RIR; when 12 reps feel like 2 RIR, add 5-10 lb. Heavier than the cable version, so it sits on the strength day.	straight-double-progression
6	Cable Cuff Rear Delt Fly (Grip-Bypass)	2 × 15-20	0-1 RIR	75	Antagonist/postural filler to balance all the anterior pressing (desk-and-running lifter). FTS cable + wrist cuffs, light — start a load you can control for 15-20 with 1 RIR. Superset these in the rest of the lateral raise or close-grip to compress session time. Not counted as a primary Push muscle — joint-health and shoulder-balance insurance.	antagonist-superset with the lateral raise / close-grip (time-efficiency)

- **Incline Dumbbell Bench Press** — *cue*: Neutral-to-45-deg angled grip, elbows ~45-60 deg from torso (not flared to 90), drive to a deep but pain-free stretch at the bottom, press up and slightly back. Control the eccentric ~2s. Keep the bar path over the upper chest, not the throat. · *accommodation*: Shoulder painless-crepitus: incline DB IS the shoulder-friendly anchor (neutral/angled grip, self-selected path) — this replaces heavy strict barbell OHP. Wrist tendinopathy: angled/semi-neutral grip keeps the wrist stacked; wrist wraps allowed. Left pinky: do NOT hang the 60-75s from a closed hand at the bottom of the rack — use the knee-kickup to load and a controlled set-down; if the pinky catches, swap to the Smith Incline (machine-loaded, grip-bypassed). TMJ: relax jaw / soft bite (or mouthguard) on the top set, no hard clench. · *cites*: exercise.claim.19-heavy-load-required-for-hypertrophy, exercise.claim.18-zero-rir-beats-low-rir-closer-is-more-growth, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, frequency, weekly_sets, exercise.constraint.mark.right-shoulder-crepitus
- **Smith Machine Flat Bench Press** — *cue*: Slight arch, shoulder blades pinned down/back, touch ~lower-sternum, elbows tucked ~45 deg. The fixed path means you can push the back-off set close to failure without a spotter. · *accommodation*: Shoulder: flat Smith is a fixed, controllable path — safer than free barbell for the crepitus shoulder; keep ROM pain-free. Wrist: wraps allowed; the bar sits in a stacked wrist. Left pinky: bar is machine-loaded so grip is bypassed. TMJ: jaw soft on the top set. · *cites*: exercise.tech.reverse-pyramid, exercise.claim.6-reverse-pyramid-more-muscle-than-straight-sets, exercise.claim.mu-incline-plus-flat-maximizes-pec, session_sets, exercise.constraint.mark.right-shoulder-crepitus
- **Close-Grip Bench Press** — *cue*: Shoulder-width grip (going too narrow loads the wrist, not more triceps), elbows tucked, touch lower chest, full lockout to load the triceps. Keep the bar over the elbows. · *accommodation*: Wrist tendinopathy: keep the grip shoulder-width (a too-narrow grip cranks wrist extension) and use wraps; the Smith keeps the wrist stacked. Left pinky: Smith bar is machine-loaded — grip bypassed. Shoulder: tucked elbows and moderate ROM keep it shoulder-friendly. TMJ: relaxed jaw on heavy sets. · *cites*: weekly_sets, exercise.claim.19-heavy-load-required-for-hypertrophy, exercise.constraint.mark.right-wrist-tendonitis, exercise.constraint.mark.left-pinky
- **Machine Lateral Raise** — *cue*: Lead with the elbow, raise to ~shoulder height in the frontal plane, brief pause, control down. Don't shrug — keep the upper trap quiet. · *accommodation*: Wrist + left pinky: machine pad/handle means grip is essentially bypassed — no DB to hang from the pinky. Shoulder: raising to ~shoulder height (not overhead) and the smooth machine arc keep it crepitus-friendly. Isolation, so taking the last set to failure is fine (failure only on isolation per the evidence rules). · *cites*: exercise.claim.mu-cable-lateral-raise-tension-at-length, exercise.claim.18-zero-rir-beats-low-rir-closer-is-more-growth, weekly_sets, frequency, exercise.constraint.mark.left-pinky
- **Overhead Barbell / EZ-Bar Triceps Extension** — *cue*: Elbows tucked and pointed forward, lower behind the head to a full long-head stretch, extend without flaring. Keep the upper arms still. · *accommodation*: Wrist tendinopathy: EZ bar's angled grip is wrist-friendlier than a straight bar — use it; wraps allowed. Left pinky: if catching the bar overhead is uncomfortable, load via the cable rope overhead version instead (grip-bypassed) — but keep the long-head stretch. TMJ: relaxed jaw. · *cites*: weekly_sets, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, exercise.constraint.mark.right-wrist-tendonitis

- **Cable Cuff Rear Delt Fly (Grip-Bypass)** — *cue*: Slight forward lean, lead with the pinky/elbow, squeeze the rear delt, control the return. Keep it strict and light. · *accommodation*: Left pinky: FTS wrist cuffs put the load on the wrist, fully bypassing the grip — this is the named grip-bypass tool. Wrist: cuff keeps the wrist neutral. Shoulder: external-rotation/horizontal-abduction pattern is prehab-friendly for the crepitus shoulder. · *cites*: exercise.tech.antagonist-superset, exercise.claim.11-supersets-compromise-hypertrophy, exercise.constraint.mark.left-pinky, exercise.constraint.mark.right-shoulder-crepitus

Tue — PULL A — Heavy / Controlled (before kickboxing PM)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
1	Weighted Pull-Up (neutral or pronated grip)	4 × 5-8	2-3 (last set 1-2)	180-240	Bodyweight first while fresh. Once you clear 8 clean reps at 2-3 RIR, add load via a plate or DB hung from a dip belt (plates in 5-lb steps; DBs to 75). Start ~+10-25 lb; double-progress 5-8 reps then +5 lb. If you cannot get 5 reps at 2-3 RIR, regress to Assisted Pull-Up (machine) or Iso-Lateral Pulldown for the same set count.	straight-double-progression (this is the heavy vertical anchor; keep gas in the tank — kickboxing is tonight, so do NOT grind to failure)
2	Barbell Bent-Over Row (with lifting straps)	4 × 6-10	2-3 (last set 1-2)	150-180	Olympic bar (45 lb) + plates in 5-lb steps; this is your heaviest horizontal pull. Start where 8 reps sit at ~3 RIR (commonly 95-155 lb) and double-progress 6-10 reps, then +5-10 lb. Keep the torso angle fixed; if lower back fatigues or krav/CNS feels pre-cooked, cap at 8 reps @3 RIR rather than chasing load.	straight-double-progression (strength-leaning; the one free-barbell row of the week, placed Tue NOT Sat to keep the spine off the post-krav day)
3	Plate-Loaded Iso-Lateral Pulldown	1 × 10-12	1-2	120	Precor/Hammer iso-lateral pulldown, selectorized or plate-loaded. One back-off set after the weighted pull-up to top the lats to 5 weekly heavy sets without more bodyweight grip load. Pick a load you control for 10-12 at 1-2 RIR; add a plate when you hit 12.	RPT-style back-off (lighter than the pull-up, more reps) — a time-efficient way to add a 5th lat set, not 'extra growth'
4	Incline Dumbbell Curl	3 × 8-12	1-2 (last set 0-1)	90-120	DBs in 5-lb steps to 75. Set the bench ~45-60°. Start where 10 reps sit at 1-2 RIR (commonly 20-35 lb/hand); double-progress 8-12 then next DB. Lengthened-biased curl prioritized over peak-contraction work for its modest measured growth edge.	straight-double-progression; take only the final set to 0-1 RIR (cheap isolation fatigue, last-set-to-failure premium)
5	EZ-Bar Curl	1 × 6-10	1-2	90	Fixed EZ bar (to 60 lb) or loaded EZ on the Olympic side in 5-lb steps. One heavier overall-biceps set to round out the heavier-rep biceps pass (4 direct biceps sets Tue total with the incline work). Start ~40-65 lb for 8 at 1-2 RIR; double-progress then +5 lb.	straight set (heavier, lower-rep biceps complement to the lengthened incline work — both heads loaded through a big-load full ROM)
6	Cable Face Pull	3 × 12-20	0-2	75-90	FTS Glide cable, rope or dual handles at upper-chest height. Light-moderate; pick a load you can pause-contract for 15 reps at 0-2 RIR, add a pin when you hit 20. Doubles as rear-delt volume AND posterior-cuff/scapular prehab for the crepitus shoulder.	straight set, pause at peak contraction (constant cable tension; trains rear delt + infraspinatus/teres minor + mid-trap in one move)
7	Farmer's Carry	2 × 30-40s walks	0-1	90	DBs to 75/hand (or loaded bar/handles). Walk 30-40 s at a load that challenges grip/trunk without form breakdown. Builds the lower-back/core priority isometrically and feeds forearm/trap volume — 2 forearm sets Tue.	loaded carry (timed, anti-lateral-flexion isometric for erectors + QL; a Pull-day way to add to the lower-back priority off the heavy-hinge day)

- **Weighted Pull-Up (neutral or pronated grip)** — *cue*: Neutral grip is the joint-friendly default; initiate by depressing the shoulder blades, drive elbows down-and-back to the lower chest, control a full overhead stretch each rep. Relax the jaw on the hard reps (no clench). · *accommodation*: Left pinky: loop FTS cuffs or lifting straps so the closed hand isn't crush-gripping a bar under bodyweight+load. Wrist tendinopathy: neutral grip keeps the wrist straight. TMJ: unclench on top sets. Shoulder crepitus: painless = no restriction; neutral grip is shoulder-friendly. · *cites*: exercise.move.pull-up-weighted, exercise.claim.mu-wide-grip-pulldown-lat-bias, rir:hypertrophy:trained:0-2, rir:strength:intermediate:3-5, exercise.constraint.mark.left-pinky, exercise.constraint.mark.tmj
- **Barbell Bent-Over Row (with lifting straps)** — *cue*: Brace, hinge to ~15-30° above horizontal, pull to the lower sternum/upper abdomen with elbows ~45°. Strap up so the back — not the grip — drives failure. Erectors work isometrically here, feeding the lower-back priority without a dedicated hinge on a Pull day. · *accommodation*: Left pinky: straps remove the crush-grip demand entirely (flagged item, grip_demand high → low with straps). Wrist tendinopathy: wraps optional. Lower-back priority: a controlled-torso strapped row loads erectors isometrically while painless — stop on any leg radiation/numbness. TMJ: relaxed jaw, no hard valsava. · *cites*: exercise.move.barbell-bent-over-row-straps, exercise.claim.mu-row-mid-back-vs-pulldown-lat, rir:strength:intermediate:3-5, exercise.constraint.mark.left-pinky, exercise.constraint.mark.lower-back
- **Plate-Loaded Iso-Lateral Pulldown** — *cue*: Chest-supported line of pull lets each lat work its own arc through a full overhead stretch to a hard contraction; let the scapula rise fully at the top for the lengthened position rows skip. · *accommodation*: Left pinky: handle is held, not hung — low grip demand; cuffs available if needed. Wrist: neutral/angled handle keeps the wrist comfortable. Counts toward the lat volume so the pull-up doesn't have to be taken deep before sport. · *cites*: exercise.move.machine-pulldown-plate-loaded, exercise.claim.6-reverse-pyramid-more-muscle-than-straight-sets, exercise.tech.reverse-pyramid, rir:hypertrophy:trained:0-2
- **Incline Dumbbell Curl** — *cue*: Let the arm hang fully behind the torso for a deep biceps stretch at the bottom — that lengthened position is the point. Supinate hard, no swing. Long head emphasis. · *accommodation*: Left pinky: 20-35 lb DBs are light enough to hold safely; if the pinky jams, switch to a Cable Curl with cuffs (same lengthened intent). Wrist: supinated DB grip is usually fine; reduce load if the wrist talks back. · *cites*: exercise.move.incline-dumbbell-curl, exercise.claim.mu-lengthened-curl-vs-preacher, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, rir:hypertrophy:intermediate:0-2-isolation, exercise.constraint.mark.left-pinky
- **EZ-Bar Curl** — *cue*: EZ camber spares the wrist vs a straight bar. Controlled tempo, no body english on a Pull day where the erectors already worked. · *accommodation*: Wrist tendinopathy: EZ bar's angled grip is the wrist-friendly choice over the straight barbell (less supination strain). Left pinky: if the closed-hand load bothers it, use FTS cuffs or swap to Machine Curl. TMJ: relaxed jaw. · *cites*: exercise.move.ez-bar-curl, exercise.constraint.mark.right-wrist-tendonitis, rir:hypertrophy:trained:0-2
- **Cable Face Pull** — *cue*: Pull to the forehead/eyes with elbows high, externally rotating at the end so the knuckles face back. Squeeze, slow eccentric. This is your rear-delt + cuff insurance that keeps the presses painless. · *accommodation*: Shoulder crepitus: face pulls are the recommended scapular-stability/external-rotation prehab for this shoulder. Wrist/pinky: rope handle is light and held loosely; no crush-grip. Pure constant-tension cable, wrist neutral. · *cites*: exercise.move.cable-face-pull, exercise.constraint.mark.right-shoulder-crepitus, rir:hypertrophy:general:0-3, exercise.move.cable-external-rotation-90-abduction
- **Farmer's Carry** — *cue*: Tall posture, ribs down, brace as if holding a plank while walking. Trains erectors/QL and grip-trap endurance together — the functional-strength + running-tightness fix. · *accommodation*: Left pinky: use FTS cuffs or straps if a closed crush-grip on the handle jams the pinky (crush-grip is deferred while symptomatic; the carry still trains everything else). Wrist: neutral grip. Lower-back priority: painless loaded carry = progress; stop on red flags. · *cites*: exercise.move.farmers-carry, exercise.constraint.mark.lower-back, exercise.constraint.mark.left-pinky, rir:hypertrophy:general:0-3

Wed — LEGS B — Heavy Hinge / Lower-Back PRIORITY → Thu REST (heavy compound, then higher-rep pump accessories)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
1	Conventional Barbell Deadlift	4 × 5	2-3 (last set 1-2)	210-300	Olympic bar with straps. Start ~245-275 lb for 4x5 @2-3 RIR. Add 5-10 lb/week while it stays PAINLESS (the priority weak-point gets the most consistent load progression). Long rests (3.5-5 min). If pulling from the floor isn't fresh, sub a Barbell Good-Morning or RDL as the hinge.	straight-sets / double-progression; this is the one place to bank a RIR buffer — keep 2-3 RIR (failure on the deadlift is slightly inferior for strength, triples acute fatigue, and threatens the priority lower back)
2	Stiff-Leg Deadlift (SLDL)	3 × 8-10	1-3	150-180	Olympic bar with straps. Start ~135-165 lb for 3x10 @2 RIR (lighter than the deadlift — this is the long-length ham/glute/erector secondary, run at higher reps for the pump pass). Double-progress reps then +5-10 lb while painless.	straight-sets / double-progression at a deliberate long-length stretch (the longer-ROM direction is corroborated by hypertrophy trials on other hinges; erector fatigue caps usable extra ROM)
3	Seated Leg Curl	3 × 15-20	0-1	75-90	Precor selectorized seated leg-curl. Higher-rep pump pass (15-20) vs Sun's 12-15 — same muscle, second weekly session split lighter/higher-rep per the load-independence of growth. Start at a 20-rep load @1 RIR; progress reps then weight.	myo-reps or a double-drop on the last set (activation set to ~1 RIR, then 3-4 short-rest mini-clusters) — time-efficient way to bank effective reps, equivalent growth at matched volume, used for time/fatigue not 'more growth'
4	Lying (Prone) Leg Curl	2 × 15-20	0-1	75-90	Precor selectorized lying leg-curl. Start at a 20-rep load @1 RIR; progress reps then weight. Programmed alongside the seated curl to cover BOTH hamstring functions/heads (the prone hip angle and knee-flexion bias hit the short head differently than the seated variant).	straight-sets to 0-1 RIR; optionally antagonist-superset with a light leg extension if quad volume feels low on Wed (quads are otherwise indirect from the deadlift here)
5	Machine Hip Thrust	3 × 12-15	1-2 (last set 0-1)	90-120	Plate-loaded hip-thrust machine. Start at a load for 3x15 @1-2 RIR; add a plate when the top of the range is clean. Very high SFR (the machine offloads the spinal setup vs barbell) — ideal after two heavy hinges when the lower back is already taxed.	straight-sets / double-progression with an optional last-set myo-rep cluster for time-efficiency; short-length glute emphasis to complement the long-length hinge loading already done today
6	Seated Calf Raise (Bent-Knee, Soleus)	3 × 15-25	0-1	60-90	No leg-press or calf machine at 290 — rig the bent-knee soleus raise seated with a loaded barbell or heavy DBs resting across the knees (toes on a block for full stretch). Start at a load for 20-25 reps @1 RIR; progress reps then load. This is the higher-rep SOLEUS pass complementing Sun's standing gastroc pass.	straight-sets to 0-1 RIR with a deep loaded stretch and bottom lengthened partials on the last set — soleus responds to high reps and stretch
7	Ab-Wheel Rollout	3 × 8-12	1-3	90-120	Bodyweight; progress by ROM (knees → toes → standing) and lever length rather than added load. Start from knees for 3x10 @2 RIR; extend reach / move toward standing as you get stronger. This is the heavy anti-extension piece of the core PRIORITY build.	straight-sets / ROM-progression (the lengthened-range anti-extension bias trains the whole rectus and braces the erectors as anti-extensors — the functional-strength fix)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
8	Farmer's Carry	3 × 30-40s walks	0-1	90	DBs up to 75 lb each (the 290 DB ceiling) or a loaded barbell/trap setup. Start with a load you can carry tall and braced for 30-40s; progress load while posture stays perfect. Loads the erectors + QL isometrically and the traps — the carry pillar of the lower-back priority engine.	straight loaded carries (timed); run as a finisher. For anti-lateral-flexion emphasis, alternate rounds as a SUITCASE carry (load one side) to hammer the QL/obliques

- **Conventional Barbell Deadlift** — *cue*: Erector + glute + hamstring PRIMARY. Neutral braced spine, lats tight, push the floor away, lock out with the glutes. Reset each rep. Stop instantly on any leg radiation / numbness / give-way / sharp catch — those are red flags, not soreness. · *accommodation*: Lower-back PRIORITY: painless = progress; the listed red flags = stop and reassess. Placed Wed so the most fatiguing hinge is followed by Thu full rest, and kept off the krav/kickboxing days. Left pinky: STRAPS bypass the grip (no closed-hand hang fail). Wrist: neutral + straps. TMJ: relaxed jaw / mouthguard on the brace, no clench. · *cites*: exercise.constraint.mark.lower-back, exercise.claim.17-failure-required-to-maximize-growth-and-strength, exercise.claim.18-zero-rir-beats-low-rir-closer-is-more-growth, exercise.constraint.mark.left-pinky, exercise.constraint.mark.tmj
- **Stiff-Leg Deadlift (SLDL)** — *cue*: Stiffer knees than the RDL, glute-max + erector at long muscle length, maximal hamstring stretch to mid-shin, hard lockout. Keep the spine neutral — back position quality is the limiter, not load. · *accommodation*: Lower-back priority — second hinge of the day, so autoregulate up a set/RIR if the deadlift was heavy. Straps bypass the pinky; neutral wrist. Stop on red flags. · *cites*: exercise.claim.ml-lengthened-hip-hinge-deficit-step-rdl, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, exercise.constraint.mark.lower-back, exercise.constraint.mark.left-pinky
- **Seated Leg Curl** — *cue*: Hip-flexed long-length curl for the hip-extensor heads. Full contraction squeeze, controlled stretch. Chase the burn here — isolation failure is cheap and captures the small hypertrophy premium. · *accommodation*: Machine — pinky/wrist/shoulder all bypassed. · *cites*: exercise.claim.ml-seated-vs-lying-leg-curl-length-bias, exercise.claim.19-heavy-load-required-for-hypertrophy, exercise.tech.myo-reps, exercise.claim.9-rest-pause-myo-reps-more-muscle-per-time
- **Lying (Prone) Leg Curl** — *cue*: Prone curl biases biceps femoris through the knee-flexion ROM. Hips pinned, no pelvis lift, full ROM both directions. · *accommodation*: Machine — fully constraint-friendly. No grip/wrist/shoulder load. · *cites*: exercise.claim.ml-rdl-hip-dominant-vs-leg-curl-knee-dominant-hamstring, exercise.claim.ml-seated-vs-lying-leg-curl-length-bias, weekly_sets
- **Machine Hip Thrust** — *cue*: Short-length peak-contraction glute work — included for the length coverage it ADDS to the squats/hinges, not as a superior glute builder. Full hip extension, posterior tilt, ribs down, pause at lockout. · *accommodation*: Machine, spine offloaded — lowest lower-back demand of the glute options, deliberately placed after the hinges. No grip/hang (pinky safe), no overhead (shoulder safe). · *cites*: exercise.claim.ml-hip-thrust-short-length-squat-long-length-glute, exercise.claim.4-hip-thrust-superior-glute-growth, exercise.tech.myo-reps, session_sets
- **Seated Calf Raise (Bent-Knee, Soleus)** — *cue*: Knee BENT = soleus bias (gastroc is slack at the knee, so seated reaches the soleus the standing raise misses). Slow full stretch at the bottom, full squeeze at the top, no bouncing. · *accommodation*: Barbell/DB rests ON the thighs (not hung from the hand) so the left pinky and wrist carry no grip load — the constraint-clean way to rig this without the absent machine. Keep wrist neutral steadying the load. · *cites*: exercise.claim.2-standing-vs-seated-calf-raise-gastroc, exercise.claim.ml-standing-vs-seated-calf-gastroc-soleus, exercise.claim.19-heavy-load-required-for-hypertrophy, exercise.constraint.mark.left-pinky
- **Ab-Wheel Rollout** — *cue*: Brace hard, posterior pelvic tilt, resist lumbar extension as you roll out to the lengthened position — do NOT let the lower back sag. This is anti-extension, the bracing skill that pairs with the erector work for running tightness. · *accommodation*: Lower-back priority done SAFELY as anti-extension (no spinal loading/compression). Moderate grip on the wheel handles — fine for the pinky; if it jams, use a loaded barbell rollout with straps instead. Stop on any lumbar red flag. · *cites*: exercise.constraint.mark.lower-back, exercise.claim.10-lengthened-partials-more-than-full-rom, weekly_sets

- **Farmer's Carry** — *cue*: Stand tall, ribs down, brace, walk smooth — the spine stays neutral under load (erector endurance), and suitcase rounds resist the side-bend (anti-lateral-flexion). This is the loaded-carry erector/core stimulus that no back-extension machine here can give. · *accommodation*: Left pinky: use FTS cable wrist cuffs or lifting straps so the closed-hand grip is fully bypassed (carries are otherwise grip-limited — straps let the erectors/QL drive). Wrist neutral. Crush-grip work (gripper/pinch/dead-hang) stays deferred while the pinky is symptomatic. · *cites*: exercise.constraint.mark.lower-back, exercise.constraint.mark.left-pinky, exercise.constraint.mark.right-wrist-tendonitis, weekly_sets

Fri — PUSH B — Pump / Metabolite (chest, delts, triceps, rotator cuff)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
1	Smith Machine Incline Press	3 × 10-15	1-2 RIR	120	Lighter, higher-rep press anchor — same muscles as Monday, different load/rep so the 2nd weekly chest pass is the high-rep half (hypertrophy is load-independent ~30-85% near failure). Smith bar ~25 lb effective; start ~95-115 effective x12 @2 RIR. Double-progress to 15 reps then add 5-10 lb. Fixed path keeps it safe near failure and quick to re-rack between sets.	straight-double-progression
2	Cable Flye (Mid)	2 × 15-20	0-1 RIR (drop on last set)	75	Constant-tension peak-contraction chest to round out the 2nd weekly chest pass (no pec-deck in the 290 gym, so cable flye is the machine-pec stand-in). Start light enough for 15-20 clean reps; on the final set run a single drop (~drop one plate, no rest, rep to failure) for time-efficient extra reps. Cable is chosen for the tension profile/comfort, not as superior to DB flye for growth.	drop-set on the final set (for time-efficiency, not more growth)
3	Cable Cuff Lateral Raise (Grip-Bypass)	4 × 15-25	0-1 RIR (myo-reps on last 1-2 sets)	75	2nd weekly lateral-delt pass — higher reps, loaded-stretch at the bottom (cable keeps tension where the DB has none). FTS cable + wrist cuffs. Start a load good for 15-20 with 1 RIR. On the last 1-2 sets run myo-reps: activation set to ~0-1 RIR, then 3-4 mini-sets of ~4-5 reps with ~15s rests to accumulate effective reps fast. Add load when the activation set clears 25.	myo-reps on the final sets (time-efficiency / fatigue management — matched-volume equivalent to straight sets)
4	Rope Triceps Pushdown	2 × 15-25	0-1 RIR (drop on last set)	75	Lateral/medial-head pump work, very high SFR. Start a load good for ~18-20 reps with 1 RIR; on the final set run a double-drop (rep to failure, strip one plate, repeat) for cheap metabolite volume in minimal time. Spread the triceps volume across the week — this plus Monday's heavy close-grip/overhead keeps total triceps sets in the >20 range that synergist triceps benefit from.	drop-set on the final set (time-efficiency)
5	Overhead Cable Rope Triceps Extension	2 × 12-20	0-2 RIR	75	Long-head stretch bias to pair with the pushdown (which keeps the long head short). Constant cable tension at the lengthened position. Start a load good for 15-18 reps with 1-2 RIR; double-progress reps to 20 then add a plate. Superset back-to-back with the rope pushdown (same station) to compress time.	straight-double-progression (optionally agonist-superset with the pushdown for density)
6	Cable External Rotation (Elbow at Side)	2 × 12-20	1-3 RIR	60	Rotator-cuff prehab for the crepitus shoulder — joint-stability insurance that keeps the presses painless, NOT hypertrophy volume. Very light: a load you control cleanly for 15-20 reps with 2-3 RIR, elbow pinned to the side, full pain-free arc. Progress slowly (this is the smallest jump on the stack). Pair with a Full-Can Raise set if you want fuller cuff/supraspinatus coverage.	straight sets (prehab — never to failure)

- **Smith Machine Incline Press** — *cue*: ~30-deg bench, elbows ~45 deg, touch upper chest, control the stretch, smooth turnaround — chase a deep pump, not a grind. Shorter rests than Monday. · *accommodation*: Shoulder: fixed Smith path + incline angle = the shoulder-friendly press; keep ROM pain-free. Wrist: stacked wrist, wraps allowed. Left pinky: machine-loaded bar, grip bypassed. TMJ: relaxed jaw (loads are submaximal here, low clench risk). · *cites*: exercise.claim.19-heavy-load-required-for-hypertrophy, frequency, weekly_sets, exercise.constraint.mark.right-shoulder-crepitus, exercise.constraint.mark.left-pinky
- **Cable Flye (Mid)** — *cue*: Soft elbows fixed, hug the cables together to full peak contraction, control the stretch. High-to-low and low-to-high handle heights are interchangeable for variety/comfort, not region-targeting. · *accommodation*: Left pinky + wrist: cable handles are low grip-demand and the wrist stays neutral; if the pinky catches, slip on the FTS wrist cuffs. Shoulder: limit the stretch to pain-free range — no forced deep stretch on the crepitus shoulder. · *cites*: exercise.tech.drop-set, exercise.claim.8-drop-sets-more-muscle, exercise.claim.13-advanced-techniques-class-vs-volume-equated-straight-sets, exercise.claim.mu-emg-swelling-predicts-regional-growth, weekly_sets
- **Cable Cuff Lateral Raise (Grip-Bypass)** — *cue*: Cuff on the wrist, lead with the elbow, control the loaded stretch at the bottom, raise to shoulder height. On myo-rep mini-sets keep form tight — stop the cluster when reps get sloppy. · *accommodation*: Left pinky: FTS wrist cuffs are the confirmed grip-bypass — load rides the wrist, the closed hand does nothing. Wrist: cuff keeps it neutral, no gripping. Shoulder: raise to ~shoulder height only; the cable's smooth arc is crepitus-friendly. · *cites*: exercise.tech.myo-reps, exercise.claim.9-rest-pause-myoreps-more-muscle-per-time, exercise.claim.mu-cable-lateral-raise-tension-at-length, frequency, weekly_sets, exercise.constraint.mark.left-pinky
- **Rope Triceps Pushdown** — *cue*: Elbows pinned to the sides (long head shortened — that's intentional, the overhead move next covers the stretch), spread the rope at lockout, full squeeze, control the return. · *accommodation*: Wrist + left pinky: rope is low grip-demand and the wrist stays neutral — friendly to both. No clench/jaw concern at these loads. · *cites*: exercise.tech.drop-set, weekly_sets, exercise.claim.13-advanced-techniques-class-vs-volume-equated-straight-sets
- **Overhead Cable Rope Triceps Extension** — *cue*: Step forward into a split stance, elbows high and forward, reach into a full overhead long-head stretch, extend without flaring the elbows. Keep upper arms still. · *accommodation*: Wrist + left pinky: rope, low grip-demand, neutral wrist — grip effectively bypassed. Shoulder: the overhead reach is loaded through the elbow (cable), not a pressing shoulder position — keep it pain-free. · *cites*: weekly_sets, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, exercise.constraint.mark.left-pinky
- **Cable External Rotation (Elbow at Side)** — *cue*: Elbow tucked to the ribs (towel under the elbow), forearm rotates out like a gate, slow and controlled, no momentum. Stop short of failure — this is insurance, not a hard set. · *accommodation*: Shoulder painless-crepitus: this is the dedicated cuff/scapular-stability work the constraint asks for on every Push/Pull — it lets the presses stay painless. Wrist + left pinky: cable handle is light and the wrist stays neutral; cuff it if the pinky catches. Keep loads light enough that grip is never the limiter. · *cites*: exercise.constraint.mark.right-shoulder-crepitus, exercise.claim.17-failure-required-to-maximize-growth-and-strength, weekly_sets

Sat — PULL B — Pump / Light Spinal Load (after krav AM)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
1	Chest-Supported Row (Machine / Plate-Loaded)	4 × 10-15	1-2 (last set 0-1)	120	Hammer/Precor chest-supported or plate-loaded row, selectorized or 5-lb plate steps. Higher-rep than Tue's barbell row. Start where 12 reps sit at 1-2 RIR; double-progress 10-15 then add a plate/pin. Torso support = near-zero lower-back demand, ideal after krav and after Wed's heavy hinge.	straight-double-progression (the Sat horizontal anchor; chest support removes the spinal/stabilizer cost on the post-krav day)

#	Exercise	Sets × Reps	RIR	Rest	Load	Structure
2	Single-Arm Cable Lat Pulldown	3 × 12-20	0-2 (last set 0-1)	90	FTS Glide cable / Precor pulldown, single handle. Per-side loading lets each lat reach an unrestricted overhead stretch. Start where 15 reps sit at 0-2 RIR; double-progress 12-20 then +1 pin. Stretch-biased vertical lat work for the lighter day.	straight set, emphasize the stretch (single-arm = bigger ROM and a deeper lengthened position than the bilateral bar)
3	Lat Prayer (Cable Straight-Arm Pullover)	1 × 12-20	0-1	75	High cable, straight arms, rope or bar. Light — this is the lengthened-lat isolation set that tops the lats to 4 weekly on the light day. Pick a load you can control to a hard contraction for 15-20 reps; add a pin at 20.	straight set / lengthened-bias finisher (loads the lat where rows lose tension — the long stretched position, matching the lat's strength curve)
4	Cable Curl	2 × 15-25	0-1	75	FTS Glide cable, straight or EZ attachment. Constant-tension high-rep biceps for the pump pass. Run the second set as a drop set for time-efficiency: hit failure ~20 reps, strip ~25%, continue. Start where 18-20 reps reach 0-1 RIR.	set 1 straight, set 2 drop set (drop sets = equivalent growth at matched volume; used here purely to compress session time after krav, not for 'more growth')
5	Rope Cable Hammer Curl	2 × 12-20	0-2 (last set 0-1)	75	FTS cable rope, neutral grip. Hits brachialis + brachioradialis (the second biceps pass spreads volume; also counts toward forearm/brachioradialis work). Start where 15 reps sit at 0-2 RIR; double-progress 12-20 then +1 pin.	straight-double-progression (neutral-grip = wrist-friendly; rounds out the 2x/week biceps + adds brachioradialis)
6	Cable Cuff Rear Delt Fly (Grip-Bypass)	3 × 15-25	0-1	60- 75	FTS Glide cables with wrist cuffs, crossed in front. Light — rear delt responds to high-rep constant tension. Run the last set as myo-reps (activation set to ~0-1 RIR, then 3-4 mini-sets of ~5 reps with 15-20 s rest) to save time. Start where 18-20 reps reach 0-1 RIR.	2 straight + 1 myo-reps final set (myo-reps = comparable growth in less time; used for time-efficiency on the pump day, not extra growth)
7	Cable Wrist Curl	2 × 20-30	0-2	60	Low FTS cable, forearms supported, wrist flexion. Lengthened-bias wrist-flexor work — the third forearm set of the week (Sat carries 3 forearm sets with the hammer curl + this). Light; train to 0-2 RIR for 20-30 reps, add a pin when you top the range.	straight set, high-rep (cable keeps tension through the lengthened bottom; finisher-style, low cost)

- **Chest-Supported Row (Machine / Plate-Loaded)** — *cue*: Pin the chest to the pad, row to the lower ribs, drive elbows back, full protraction stretch at the front. No torso heave — the pad does the stabilizing so the mid-back drives failure. · *accommodation*: Lower-back priority: chest-supported = minimal erector fatigue, deliberately chosen because Wed already loaded the spine hard and krav was this AM. Left pinky: handles held, not hung; cuffs available. Wrist: neutral handle. Autoregulate — if krav was brutal, drop to 3 sets or +1 RIR. · *cites*: exercise.move.chest-supported-row-machine, exercise.claim.mu-row-mid-back-vs-pulldown-lat, exercise.constraint.mark.lower-back, rir:hypertrophy:trained:0-2
- **Single-Arm Cable Lat Pulldown** — *cue*: Reach tall to let the scapula fully elevate/protract at the top, then pull the elbow to the hip pocket. Pause the stretch. Lengthened-position bias for the modest growth edge — not for 'region targeting,' which is refuted. · *accommodation*: Left pinky: cable handle held lightly with optional cuff; no bodyweight crush-grip (unlike a pull-up), so this is the pinky-friendly vertical option for the lighter day. Wrist: neutral handle. · *cites*: exercise.move.lat-pulldown-single-arm, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, exercise.claim.0-grip-orientation-regional-lat-targeting, rir:hypertrophy:general:0-1-light, exercise.constraint.mark.left-pinky
- **Lat Prayer (Cable Straight-Arm Pullover)** — *cue*: Hinge slightly, sweep straight arms from overhead down to the thighs, feeling the lats stretch overhead and fully contract at the bottom. Stretch-biased lat add-on (a defensible heuristic, not 'rows are useless'). · *accommodation*: Left pinky/wrist: low grip demand, neutral wrist, fully constant-tension cable — one of the

most joint-friendly lat moves available. No spinal load on the post-krav day. · *cites*: exercise.move.lat-prayer, exercise.claim.mu-row-lat-tension-vs-pulldown-leverage, exercise.claim.ml-lengthened-bias-grows-more-but-not-stretch-mediated, rir:hypertrophy:general:0-1-light

- **Cable Curl** — *cue*: Constant cable tension top-to-bottom, no rest at lockout, squeeze each rep. High-rep at light load → train to 0-1 RIR so high-threshold units still fire. · *accommodation*: Left pinky: cable + cuff fully bypasses the grip (confirmed FTS workaround) — the reason cable beats hanging DBs for the symptomatic pinky. Wrist: neutral/EZ attachment keeps it comfortable. · *cites*: exercise.move.cable-curl, exercise.tech.drop-set, exercise.claim.8-drop-sets-more-muscle, rir:hypertrophy:general:0-1-light, exercise.constraint.mark.left-pinky
- **Rope Cable Hammer Curl** — *cue*: Neutral hammer path, keep the elbows pinned, drive the thumb toward the shoulder. Builds the brachialis under the biceps and the brachioradialis of the forearm in one move. · *accommodation*: Wrist tendinopathy: neutral grip is the recommended wrist-sparing variant. Left pinky: rope held loosely, cable-loaded, optional cuff — no crush-grip. Pure constant tension. · *cites*: exercise.move.rope-cable-hammer-curl, exercise.constraint.mark.right-wrist-tendonitis, rir:hypertrophy:general:0-3, exercise.constraint.mark.left-pinky
- **Cable Cuff Rear Delt Fly (Grip-Bypass)** — *cue*: Wrist cuffs let you forget the hands entirely and just feel the rear delt sweep open. Lead with the elbows, pause at full horizontal abduction. Rear-delt volume + posture work for the desk-and-running lifter. · *accommodation*: Left pinky: cuffs are the explicit confirmed FTS bypass — zero grip demand. Wrist: cuff means no wrist load at all. Shoulder crepitus: rear-delt/scapular work is part of keeping the shoulder healthy. · *cites*: exercise.move.cable-cuff-rear-delt-fly, exercise.tech.myo-reps, exercise.claim.13-advanced-techniques-class-vs-volume-equated-straight-sets, exercise.constraint.mark.left-pinky, rir:hypertrophy:general:0-1-light
- **Cable Wrist Curl** — *cue*: Let the cable pull the wrist into full extension at the bottom for the stretch, then curl up. Keep the forearms locked on the bench/thigh so only the wrist moves. · *accommodation*: Wrist tendinopathy: cable lets the wrist move through a controlled, low-load arc — back off range/load if it's symptomatic (it's currently quiet). Left pinky: cable/low grip demand; finger-rolling and crush-grip (gripper/pinch/dead-hang) are DEFERRED while the pinky is symptomatic — this open wrist-flexion move is safe. · *cites*: exercise.move.cable-wrist-curl, exercise.constraint.mark.right-wrist-tendonitis, exercise.constraint.mark.left-pinky, rir:hypertrophy:general:0-3

Dynamic mesocycle (the progression)

Length: 10 weeks total = two stacked 4-week accumulation blocks + 1 deload after each (4+1, 4+1). Runs as one continuous mesocycle on top of the static PPL x2 (Sun Legs / Mon Push-heavy / Tue Pull+kickboxing / Wed Legs-heavy-hinge / Thu REST / Fri Push-pump / Sat Pull-lighter+krav). 10 wks is the floor for the Martikainen descending-RIR design (two 5-wk waves incl. deload) and lets the erector/core priority bank ~8 consecutive loaded weeks before the 8-12wk re-check; extend to 12 (three short blocks) only if joints/recovery are still rising at wk10.

Weeks	Focus	Volume	Intensity
Block A · Wk1-2 (MEV entry / re-acclimat e)	Re-groove every fixed-core compound at the new block's working weights after the prior deload. Hip-hinge erector/core PRIORITY already seeded at the TOP of its range. Fixed-core anchors: Wed heavy hinge = conventional or trap-bar deadlift (trap-bar = lower erector shear, claim regional_bias) cycling with RDL/SLDL; Mon heavy press = incline DB or machine/Smith incline press (shoulder-crepitus: NO heavy strict barbell OHP — constraint.mark.right-shoulder-crepitus); Sun squat = high-bar/front/Smith squat to depth (ml-deep-squat-more-leg-growth). Pull anchor = chest-supported / machine / cable rows + pulldown (frequency is just the volume vehicle — claim.16, loading frequency rows).	~10-12 hard sets/muscle/wk = MEV (weekly_sets intermediate MEV 8-12; RP MEV = mesocycle start). PRIORITY erectors/core enter at the HIGH end ~12-14 (split across Wed hinge + Sun/Sat carries + core slot) since painless+trainable (constraint.mark.lower-back). Per-session cap ~6-8 quality sets/muscle (session_sets hypertrophy 6-8 ceiling) — PPL x2 spreads the rest. ~6-9 weekly sets per smaller muscle.	RIR HIGH end of each block's wave. Heavy compounds (squat/deadlift/hinge/fixed-core press+pull) 3-4 RIR — strength is preserved across a wide RIR band and 0 RIR is slightly INFERIOR for squat/bench/DL (claim.17, claim.18, rir-strength intermediate 3-5 row: Robinson 2025/Ruple 2023). Isolation/machine/cable 2-3 RIR, no failure yet. Deadlift never past 2-3 RIR all block (protects the priority lower back). Double progression: add reps in range first, then +5 lb / next DB once top of range hit (tech.straight-sets).
Block A · Wk3-4 (MAV ramp / intensify)	Same exercise pool held constant within the block (let the lifts express load progression — strength is a skill, claim.16/frequency-strength). Push the erector/core load most aggressively here: add reps/weight to RDL/SLDL/good-morning and loaded carries every week while painless. Pump-day advanced techniques switch ON (Fri Push, Sat Pull only): drop sets / rest-pause / myo-reps on cable & machine isolations purely for time + fatigue-economy, NOT extra growth (claim.13, tech.drop-set/rest-pause/myo-reps — all volume-equated-equivalent).	Ramp +1-3 sets/muscle/wk: ~14 → ~17-18 hard sets/muscle by wk4 (weekly_sets MAV 12-20, +1-3/wk ramp). Erectors/core ride toward ~16-18 (its MAV ceiling). Most muscles touch ~18 (approaching MRV ~18-25) only by the final loaded week. Added volume lands on the pump days via the advanced techniques so per-session time stays bounded.	RIR walks DOWN across the block (descending-RIR = equal gains at LOWER average RPE, Martikainen 2025, rir-hypertrophy intermediate descending row). Heavy compounds 2-3 RIR now (last set of a non-deadlift press/squat may touch 1 RIR by wk4); deadlift stays 2-3 RIR, never failure. Isolation 0-2 RIR, final set of each isolation taken to failure (cheap fatigue, captures the small hypertrophy premium — rir-hypertrophy 0-2 isolation row). Light high-rep cable/calf work 0-1 RIR so high-threshold units recruit at light load (rir-hypertrophy 0-1 light-load row, load_pct_1rm 30-floor).
Wk5 (DELOAD 1)	Fatigue-shed, not detrain. Keep the movement patterns, drop the dose. Re-test nothing; just move well. Deadlift to a token top-set only or swap to trap-bar at light load to spare the erectors.	~half volume — drop to MV ~6 sets/muscle/wk (weekly_sets MV 6 maintenance row; RP deload to MV). Cut sets, not the exercises. Advanced techniques OFF.	3-6 RIR all sets, no failure anywhere (rir-hypertrophy intermediate deload 3-6 row: directly removes the fatigue-generating last reps). Loads ~60-70% of block top sets.
Block B · Wk6-7 (MEV+ re-entry, rotated)	RESTART at MEV with the dynamic-accessory lifts ROTATED (see accessory_rotation) and fixed-core vari/grip rotated for joint health (e.g. trap-bar↔conventional↔RDL as the Wed hinge; incline DB↔machine incline↔Smith incline as the Mon press). Block B starts slightly heavier than Block A wk1 (new rep/load PRs carried over via double progression). Erector/core priority re-seeded at the top of range again.	Re-enter ~11-13 sets/muscle (MEV), erectors/core ~13-15. Slightly above Block A's entry because tolerance rose (MAV note: tolerance climbs with accumulated training).	Reset RIR to the high end: heavy compounds 3-4 RIR, isolation 2-3 RIR. Re-establish the descending-RIR runway. Deadlift family again capped 2-3 RIR.

Weeks	Focus	Volume	Intensity
Block B · Wk8-9 (MAV ramp / peak loaded volume)	Highest training stress of the mesocycle. Rotated accessories now loaded hard; pump-day advanced techniques ON again. This is where the erector/core weak-point gets its most consistent weekly load progression (still only while painless). Optional RPT intensification on 1-2 fixed-core compounds (see rpt_protocol) for time-efficient heavy exposure.	Ramp to ~18-20 sets/muscle by wk9 (MAV top / brushing MRV 18-25). Erectors/core at their ~18-20 ceiling. Briefly touching MRV is fine; if wk9 performance can't match wk8 at equal rest, that muscle has hit MRV — hold or cut a set (MRV row: back off when performance drops).	Descend to the floor: heavy non-deadlift compounds 1-2 RIR (reserve 0-1 RIR for the very LAST set only — claim.18 / rir-hypertrophy trained 'last set only' row). Deadlift still 2-3 RIR, NEVER 0 — failure on the big barbell lifts ~triples acute neuromuscular fatigue and is slightly inferior for strength (rir-hypertrophy general 'failure ~3x fatigue' row, claim.17). Isolation 0-1 RIR, last set to failure.
Wk10 (DELOAD 2 + 8-12wk RE-CHECK)	Deload as wk5, AND run the running re-check: re-read goals/constraints, log which muscles hit MRV early (candidates to cap next cycle), confirm the lower back stayed painless across all 8 loaded weeks, audit shoulder/wrist/pinky status, decide whether to extend to a 3rd short block (wk11-12) or restart a fresh mesocycle.	MV ~6 sets/muscle, half the lifts cut. Techniques OFF.	3-6 RIR, no failure. Then re-plan.

Reverse-pyramid (fixed cores): Reverse-pyramid is applied ONLY to the fixed-core compounds (squat, deadlift/hinge, fixed-core incline press, fixed-core row) and ONLY as a loading-organization / time-efficiency tool — it is NOT more growth than equal-volume straight sets (claim.6 'reverse-pyramid more muscle' = equivalent; tech.reverse-pyramid: edge is order/freshness, not hypertrophy). Protocol: after warm-ups, the HEAVIEST top set is performed FIRST while fully fresh (claim.6 rationale: max tension/recruitment on the fresh set, unlike crescent pyramids), then load is dropped ~5-10% per back-off set to extend reps. Concrete on a fixed-core compound: Set1 = top set 4-6 reps @ the strength-leaning RIR for that block week (3-4 RIR early block, 2-3 RIR late; deadlift NEVER below 2-3 RIR — claim.17, rir-strength 3-5 row); Set2 = -7-10% for 6-8 reps @ 2 RIR; Set3 = -7-10% again for 8-10 reps @ 1-2 RIR. Use RPT on at most 1-2 fixed-core lifts per session so total fresh-heavy exposure stays manageable; the rest of the program runs straight sets with double progression (tech.straight-sets). RPT is the heavy-day vehicle (Mon heavy press, Wed heavy hinge, Sun squat); the pump days (Fri/Sat) use straight sets + advanced techniques instead. Load ceilings from the 290 algebra: Olympic bar to ~570 lb and Smith to ~550 give effectively unlimited barbell top-set room; DBs cap at 75 lb (load_algebra_290), so DB fixed-core presses will out-run the top end into higher reps — when a DB lift tops its range at 75 lb, push reps/tempo or migrate that slot to barbell/Smith/machine rather than chasing load. Wrist tendinopathy + left pinky: use neutral grip + FTS cuffs/straps on the heavy top sets (constraint.mark.right-wrist-tendonitis, constraint.mark.left-pinky); TMJ: relaxed jaw / mouthguard on the heavy top set's valsalva (constraint.mark.tmj).

Accessory rotation: Rotate the DYNAMIC-ACCESSORY lifts every block (~every 4-6 wks, at each deload boundary wk5 and wk10) for regional variety, joint health, and stimulus novelty; KEEP the fixed-core anchors stable within a block (only vary their grip/implement). Rotation is justified by joint-variety + ROM/length coverage, NOT by EMG/pump 'targeting' (emg-swelling-predicts-regional-growth = REFUTED). Pools (all from menu.json, dynamic-accessory role): BICEPS — Block A incline-dumbbell-curl (long-head stretch, mu-lengthened-curl-vs-preacher SUPPORTED) → Block B cable-curl/spider-curl (constant tension, drop-set friendly). TRICEPS — A overhead-cable-rope-extension (long-head stretch) → B rope-pushdown + skullcrusher. HAMSTRINGS split BOTH patterns every block (ml-rdl-hip-dominant-vs-leg-curl-knee-dominant SUPPORTED — program a hinge AND a curl): A = step/dumbbell-RDL + seated-leg-curl (hip-extensor heads at length, ml-seated-vs-lying-leg-curl SUPPORTED) → B = single-leg-RDL/good-morning + lying-leg-curl. QUADS — A leg-extension-extended-hip (reclined RF bias, ml-leg-extension-extended-hip-rectus-femoris SUPPORTED) + Bulgarian split squat → B sissy-squat/reverse-nordic

(lengthened RF) + walking-lunge. CALVES — alternate standing (gastroc, ml-standing-vs-seated-calf SUPPORTED) machine/Smith/dumbbell variants block-to-block + ALWAYS keep one seated/bent-knee leg-press calf raise for soleus; add standing-calf lengthened-partials in Block B. DELTS — lateral: A dumbbell/lean-away-lateral-raise → B cable-cuff-lateral-raise (FTS grip-bypass for pinky) / machine-lateral-raise; rear: face-pull ↔ cable-cuff-rear-delt-fly; rotator-cuff + scapular work (cable-external-rotation, full-can-raise, serratus-punch) every Push AND Pull, NON-rotating (constraint.mark.right-shoulder-crepitus). BACK accessories — A chest-supported/incline-bench DB row + wide-grip pulldown → B seated-cable-row-neutral + lat-prayer/straight-arm-pulldown (long-length lat work; mu-row-vs-pulldown bias is only an activation heuristic, not differential growth — REFUTED, so rotate for variety not 'targeting'). GLUTES — hip-thrust variants (machine/Smith/American) ↔ kickbacks + abduction, IN for the short-length emphasis they add, NOT as a squat replacement (ml-hip-thrust-vs-squat-glute-growth REFUTED; ml-hip-thrust-short-length complementary). CORE/ERECTOR (priority, rotates but never drops below its high range): A ab-wheel-rollout (anti-ext) + suitcase-carry (anti-lateral-flexion) + cable-crunch → B weighted-plank/captains-chair-leg-raise + farmers-carry + cable-side-bend; erector loading itself runs through the rotating hinge anchors + good-morning since NO 45-deg/GHD back-ext machine exists (equipment_notes: absent). Grip-loaded accessories (rows, carries, RDLs) ALWAYS use FTS cuffs/straps (constraint.mark.left-pinky); defer crush-grip work (gripper/pinch/dead-hang) until the pinky settles.

Deload: Cadence: a deload week every 5th week — wk5 and wk10 — i.e. after each 4-week accumulation block (the descending-RIR block ends near the fatigue floor, so deload is scheduled, not reactive). HOW: cut to ~half volume = MV ~6 hard sets/muscle/wk (weekly_sets MV 6 maintenance row; RP deload-to-MV) by dropping sets and removing the back-end accessories, while KEEPING the movement patterns so skill/load is retained. Hold ALL sets at 3-6 RIR with NO sets to failure anywhere (rir-hypertrophy intermediate deload 3-6 row — raising RIR directly removes the fatigue-generating last reps that Refalo showed cost ~3x neuromuscular fatigue and outlast 24h). Loads ~60-70% of the block's top sets. Advanced techniques (drop/rest-pause/myo-reps) OFF during deload (they are high/moderate fatigue_cost, tech rows). The deadlift is deloaded hardest — token light top-set or trap-bar only — to fully unload the priority erectors. EARLY/REACTIVE deload trigger (insert off-schedule): if you can't match the prior week's performance on a lift despite full rest, that's MRV (weekly_sets MRV row) — take that week as a deload or restart at MEV. Each deload also doubles as the autoregulation reset (re-baseline RIR for the next block) and wk10's deload carries the 8-12wk re-check.

Autoregulation (for the variable sport load + lower-back): - SPORTS ARE NOT TRAINING VOLUME — the lifting stands alone; on heavy-sport-overlap weeks you dial the LIFTING DOWN rather than leaning on the sport for stimulus (philosophy: self-sufficient). Levers, in order: (1) +1-2 RIR on the affected session, (2) drop the last 1-2 accessory sets, (3) keep big lifts at 2-3 RIR and never to failure on sport-adjacent days. - TUESDAY (Pull, kickboxing PM): Pull is deliberately on the kickboxing day so legs/push power stay fresh for the sport (scheduling_rules). Keep Tue compound rows at 2-3 RIR, never to failure; if kickboxing was brutal last week or you feel pre-fatigued, drop the last 1-2 isolation sets — do NOT add volume on top into overreaching. - SATURDAY (Pull-lighter, krav AM): Sat is the lighter chest-supported/machine Pull AFTER krav (scheduling_rules: Sat lighter). Run it +1 RIR vs a fresh day, lean on machine/cable + advanced techniques (low skill, low systemic cost). If krav was unusually hard, cut Sat to MV-ish (cut sets) or skip the pump finishers — never make it up by piling onto Sun/Mon. - WEDNESDAY heavy hinge is islanded before Thursday REST (scheduling_rules) and kept clear of krav/kickboxing days — protect this slot. If the week's sport load was high, hold the hinge at 2-3 RIR and trim accessory hinge volume, but keep the erector/core PRIORITY dose at the high end of its range whenever painless (constraint.mark.lower-back: priority build, not avoidance). - PICKLEBALL / variable extra sport: treat as added systemic fatigue, not stimulus. If a given week stacks pickleball onto kickboxing+krav, raise RIR by 1 across that week's heavy compounds and let the deload absorb the rest — missing lifting volume is made up later in the week or simply skipped, never added on top. - WITHIN-BLOCK RIR

AUTOREGULATION: ride the descending-RIR wave (start ~3-4 RIR, finish ~1 RIR on non-deadlift compounds and isolation) — matched gains at LOWER average RPE (Martikainen, rir-hypertrophy intermediate descending row). Heavy compounds 3-5 RIR early → 2-3 late, last set may touch 0-1 RIR ONLY on non-deadlift presses/squats (claim.18). Isolation 0-2 RIR, final set to failure. Light high-rep cable/calf 0-1 RIR (load-dependent failure requirement, rir-hypertrophy 0-1 light-load row). - DEADLIFT / heavy hinge NEVER to momentary failure — 0 RIR ~triples acute neuromuscular fatigue, is slightly inferior for strength, and on the deadlift it threatens the priority lower back (claim.17, claim.18, rir-strength 3-5 row: Robinson/Ruple). Cap it 2-3 RIR every set, all block. - LOWER-BACK MONITORING (priority weak-point, painless=continue): the erector/core gets the HIGH end of its volume range and the most consistent weekly load progression — but ONLY while it stays painless. STOP and reassess on any red flag (constraint.mark.lower-back red_flags): pain radiating into buttock/leg, numbness/tingling, leg weakness/foot-drop, bladder/bowel changes, night or rest pain, sharp catching/give-way. Any of these = pull hinge load back to pain-free ROM/load, swap to trap-bar/supported variants, and re-check before re-loading. - CONSTRAINTS ALWAYS ON: shoulder painless-crepitus → neutral-grip DB / machine / Smith incline are the press anchors, NO heavy strict barbell OHP, rotator-cuff+scapular work every Push/Pull (get assessed before any heavy strict overhead) — constraint.mark.right-shoulder-crepitus. Wrist tendinopathy → neutral grip / wraps / cable-machine where wrist stays neutral — constraint.mark.right-wrist-tendonitis. Left pinky (temporary) → FTS cuffs/straps/machine on every grip-loaded pull/hinge/carry; defer crush-grip — constraint.mark.left-pinky. TMJ → relaxed jaw / mouthguard, no hard clench on heavy/valsava sets — constraint.mark.tmj. - RUNNING 8-12wk RE-CHECK (at wk10 deload): re-read goals + constraints; log which muscles hit MRV early (cap them next cycle); confirm lower back stayed painless across all 8 loaded weeks; audit shoulder/wrist/pinky; decide extend-to-12 (3rd short block) vs restart at MEV. Advanced techniques stay confined to the Fri/Sat pump days for time-efficiency/fatigue-management only — never as 'more growth' (claim.13: equivalent at matched volume).

Constraint accommodations baked in

- **Left pinky (temporary):** grip-loaded moves use FTS cuffs / straps; no hanging DBs from the closed hand. Toggle off when it resolves.
- **Right shoulder (painless crepitus):** shoulder-friendly presses (neutral-grip DB / landmine / Smith / machine) instead of heavy strict barbell OHP; rotator-cuff prehab included.
- **Right wrist (quiet tendinopathy):** neutral grips / wraps; cable & machine where the wrist stays neutral.
- **TMJ:** keep the jaw relaxed / mouthguard on heavy sets.
- **Lower back (priority build):** erectors loaded via deadlift/good-morning/RDL + carries (no GHD at this gym); heavy hinge sits Wed→Thu-rest, off the kickboxing/krav days. Painless = progress; re-check running at 8–12 weeks; physio assessment still recommended.